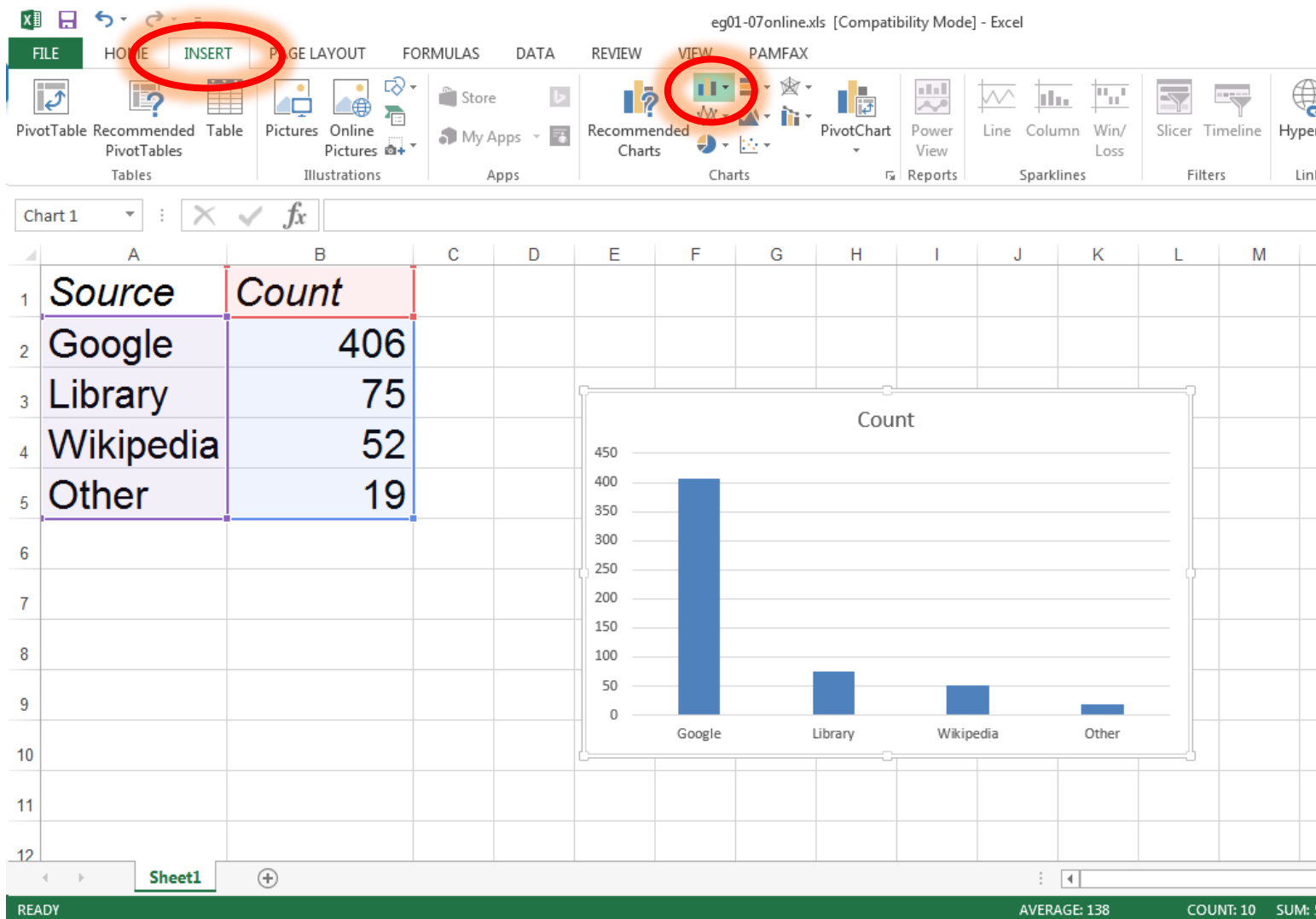
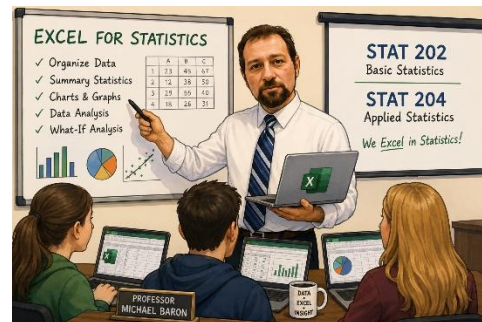
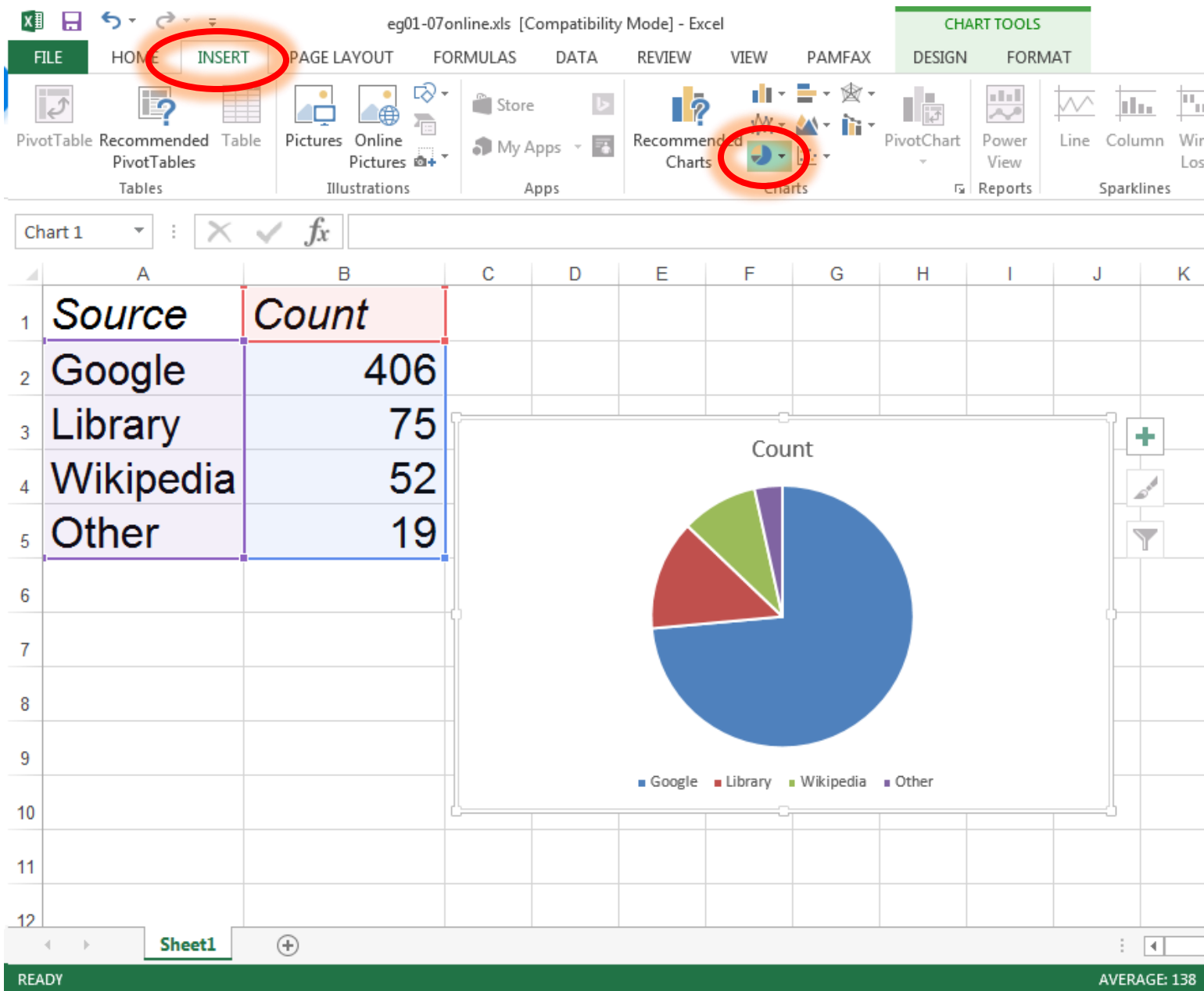


Displaying categorical data – bar charts and pie charts

Case 1. Count or percent is given for each category.

Highlight the table, names and counts of each category, then go to the “Insert” tab and find the “Charts” section. Press on a Bar Chart or a Pie Chart. You can find them in “Recommended charts” as well.





Case 2. Only raw data are given.

Calculate the counts using Excel function `=countif(range,criteria)`. It counts how many values in the chosen range meet the specified criteria.

For the range, highlight the column with the raw data. For the criterion, write the code of the corresponding category.

In the following example of Titanic passengers data, here is how you can draw pie charts of the passengers' class, gender, and survival.

eg01-36titanic.xls [Compatibility Mode] - Excel

FILE HOME INSERT PAGE LAYOUT FORMULAS DATA REVIEW VIEW PAMFAX

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SUM : =COUNTIF(A:A,F2)

	A	B	C	D	E	F	G	H	I	J
1	pclass	survived	gender	age		Pclass	Count			
2	1	1	female	29		1	=COUNTIF(A:A,F2)			
3	1	1	male	0.9167		2				
4	1	0	female	2		3				
5	1	0	male	30						
6	1	0	female	25		Survived	Count			
7	1	1	male	48		0				
8	1	1	female	63		1				
9	1	0	male	39						
10	1	1	female	53						
11	1	0	male	71						
12	1	0	male	47						
13	1	1	female	18						
14	1	1	female	24						
15	1	1	female	26						
16	1	1	male	80						
17	1	0	male							
18	1	0	male	24						
19	1	1	female	50						

Sheet1

EDIT

Screenshots Chapter 1 eg01-36titanic.xls ... 1 - Graphical displ...

To get a count of passengers in the 1st class, you can write 1 as a criterion =countif(A:A,1). Instead, as you see, I wrote "F2". Excel knows to take the value in the F2 cell, which is 1.

This way, I can extend my formula to the 2nd and the 3rd classes without rewriting the formula. Put the cursor into the left-bottom corner of the cell until the little cross becomes black and drag it to the other cells.

eg01-36titanic.xls [Compatibility Mode] - Excel

FILE HOME INSERT PAGE LAYOUT FORMULAS DATA REVIEW VIEW PAMFAX

Clipboard Font Alignment Number Styles

General

Conditional Formatting Table Styles

G2 : $\times$ $\checkmark$ fx =COUNTIF(A:A,F2)

	A	B	C	D	E	F	G	H	I	J
1	pclass	survived	gender	age		Pclass	Count			
2	1	1	female	29		1	323			
3	1	1	male	0.9167		2	277			
4	1	0	female	2		3	709			
5	1	0	male	30						
6	1	0	female	25		Survived	Count			
7	1	1	male	48		0				
8	1	1	female	63		1				
9	1	0	male	39						
10	1	1	female	53						
11	1	0	male	71						
12	1	0	male	47						
13	1	1	female	18						
14	1	1	female	24						
15	1	1	female	26						
16	1	1	male	80						
17	1	0	male							
18	1	0	male	24						
19	1	1	female	50						

Sheet1

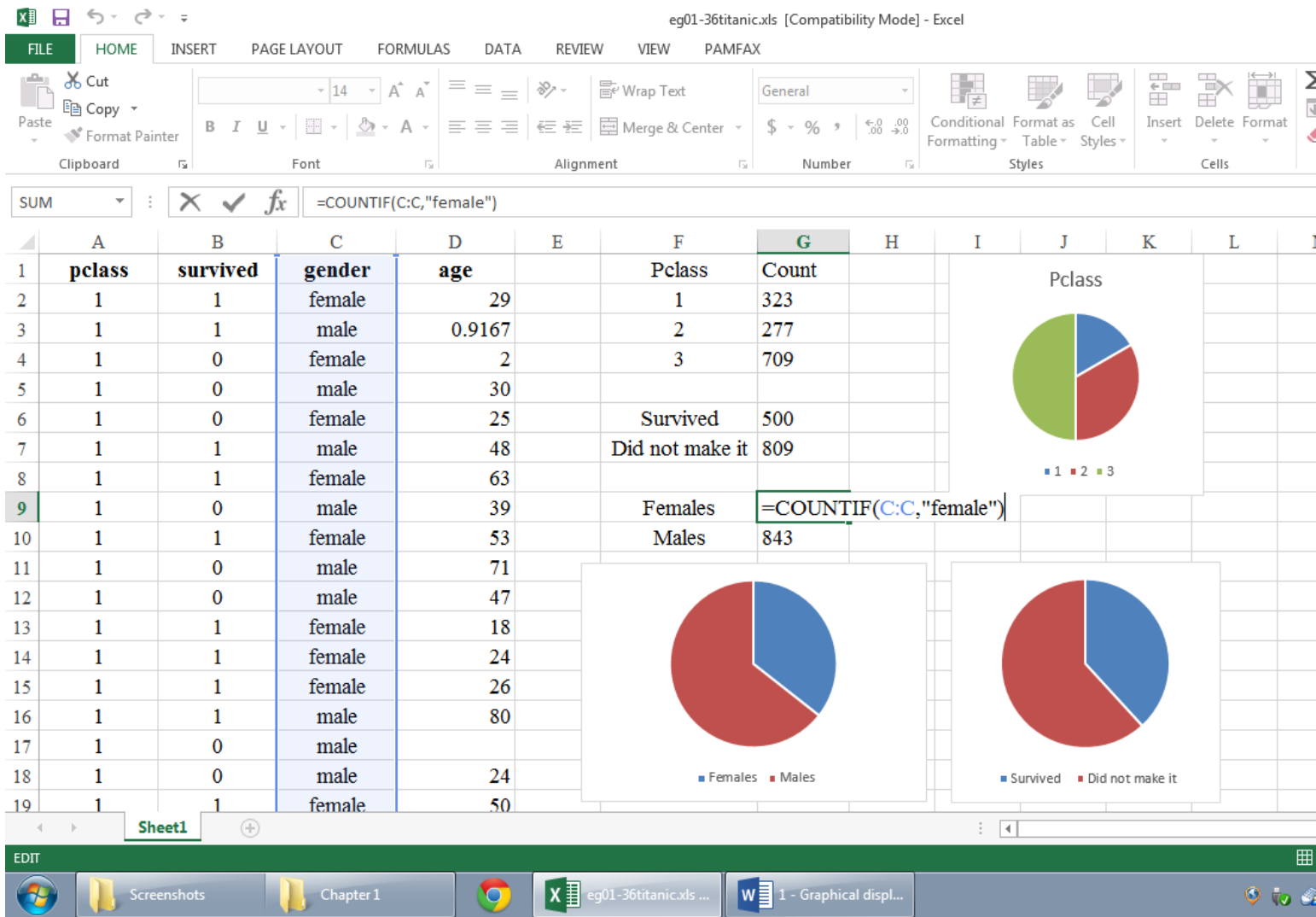
READY AVERAGE: 436.333

Screenshots Chapter 1 eg01-36titanic.xls ... 1 - Graphical displ...

After this prep, the data are ready for graphical displays. For each variable, go

Insert -> Charts -> pie chart or bar chart.

When you write criteria in =countif for the string variable “gender” (values are not numbers but strings), put values “female” and “male” into double quotes:



Displaying a relation between two variables

To see how the chance of survival on the Titanic depended on the passenger's class, we can use side-by-side bar charts.

First, we need to create a table of cross-counts. One way to do this is to sort all the data by survival. Go to the DATA tab, click Sort, and a box will appear. Choose sorting by "survived" and order from "smallest to largest" ("largest to smallest" is fine too). Click OK.

eg01-36titanic.xls [Compatibility Mode] - Excel

FILE HOME INSERT PAGE LAYOUT FORMULAS DATA REVIEW VIEW PAMFAX

From Access From Web From Text From Other Sources Existing Connections Refresh All Properties Edit Links Connections Sort Filter Sort & Filter Clear Reapply Advanced Text to Columns Flash Fill Remove Duplicates Data Validation Consolidate What-If Analysis Relationships Group Ungroup Outline

A2 1

	A	B	C	D
1	pclass	survived	gender	age
2	1	1	female	29
3	1	1	male	0.9167
4	1	0	female	2
5	1	0	male	30
6	1	0	female	25
7	1	1	male	48
8	1	1	female	63
9	1	0	male	39
10	1	1	female	53
11	1	0	male	71
12	1	0	male	47
13	1	1	female	18
14	1	1	female	24
15	1	1	female	26
16	1	1	male	80
17	1	0	male	
18	1	0	male	24
19	1	1	female	50

Sheet1 Sheet2 Sheet3

READY AVERAGE: 9.486808597 COUNT: 4977 SUM: 34759.6667

Screenshots Chapter 1 eg01-36titanic.xls 1 - Graphical displ...

As a result, all perished passengers will be listed first, followed by the survivors. Then we can apply =countif to the victims and survivors separately, as on this picture.

eg01-36titanic.xls [Compatibility Mode] - Excel

FILE HOME INSERT PAGE LAYOUT FORMULAS DATA REVIEW VIEW PAMFAX

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SUM : $\times$ $\checkmark$ fx =COUNTIF(\$A811:\$A1310,J\$1)

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	pclass	survived	gender	age			PCLASS	1	2	3			
2	1	0	female	2			Survived	123	158	528			
3	1	0	male	30			Did not make it	200	119	=COUNTIF(\$A811:\$A1310,J\$1)			
4	1	0	female	25						COUNTIF(range, criteria)			
5	1	0	male	39									
6	1	0	male	71									
7	1	0	male	47									
8	1	0	male										
9	1	0	male	24									
10	1	0	male	36									
11	1	0	male	25									
12	1	0	male	45									
13	1	0	male	42									
14	1	0	male	41									
15	1	0	male	48									
16	1	0	male										
17	1	0	male	45									
18	1	0	male										
19	1	0	male	22									

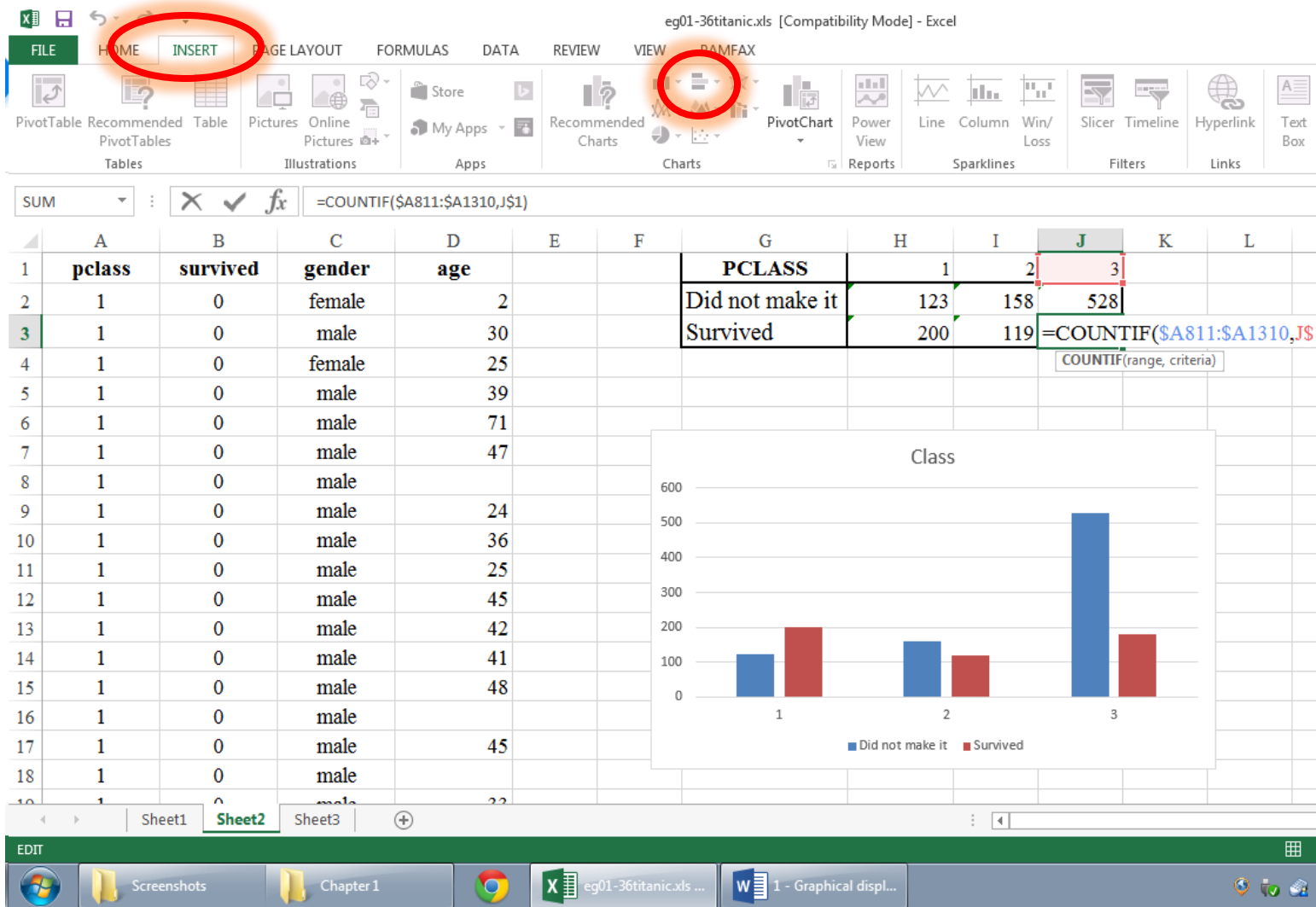
Sheet1 Sheet2 Sheet3

EDIT

Screenshots Chapter 1

eg01-36titanic.xls ... 1 - Graphical displ...

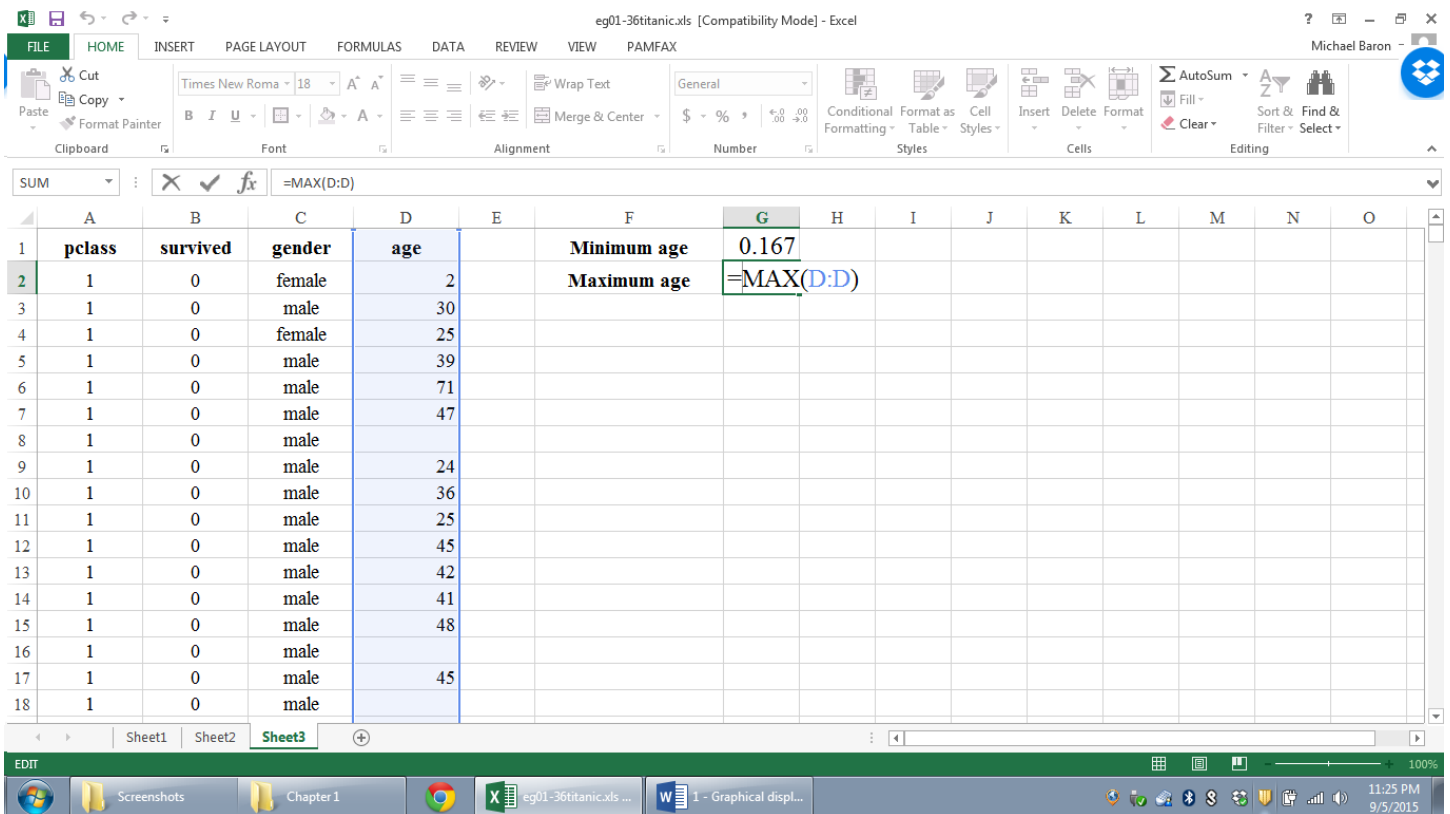
Insert -> Charts -> Bar chart produces this.



These side-by-side bar charts show that the proportion of survivors was much smaller in the 3rd class... as we saw in the movie *Titanic*.

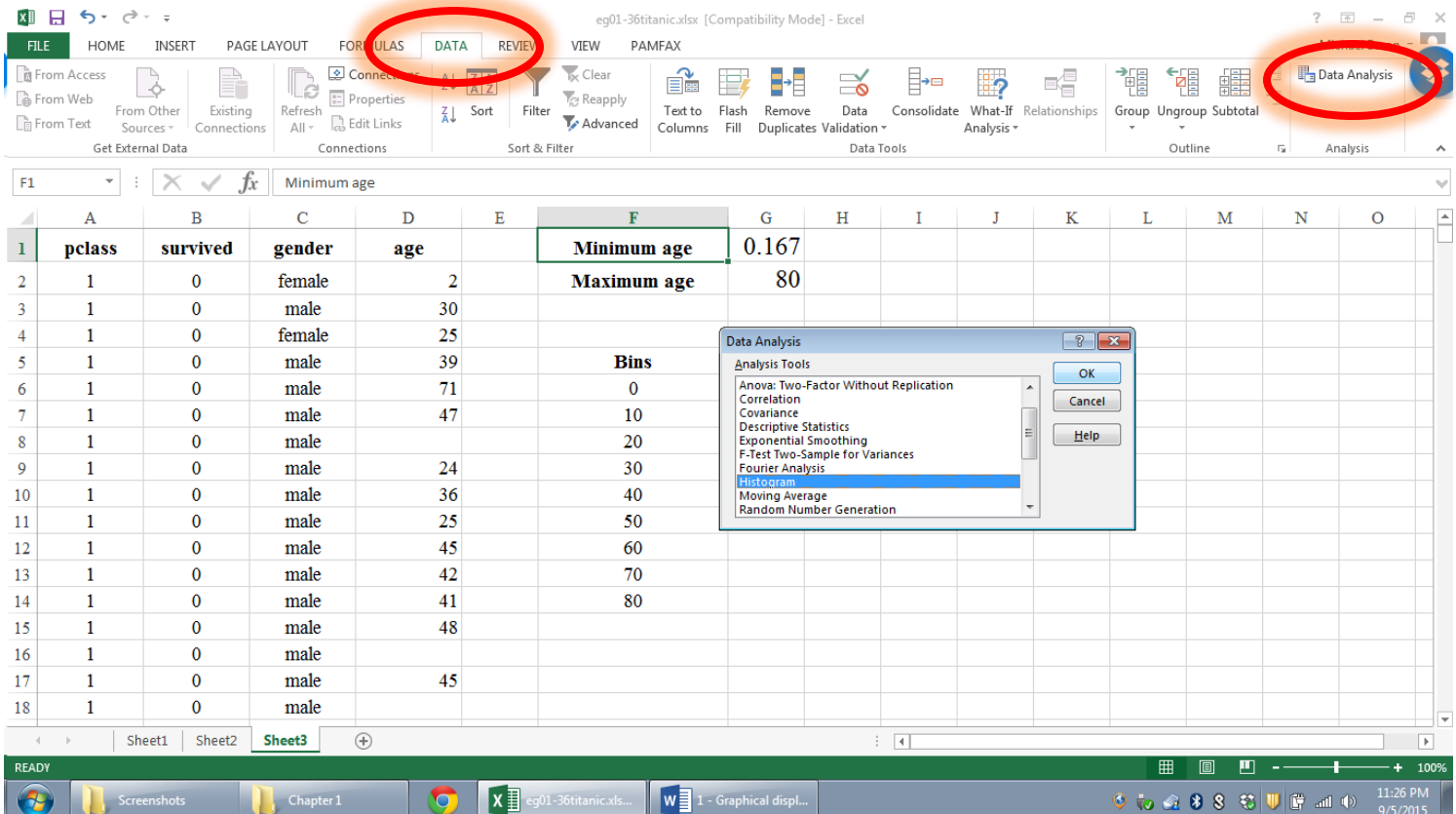
3. Displaying numerical data with histograms

Find the range of possible values of our data. If it is not known in advance, you can use =min and =max functions.



The bins can then be written manually between these min and max. Normally, arrange between 7 and 15 bins of equal width.

To produce a histogram, use the Data Analysis Toolpak. Go to the DATA tab, find the Toolpak on the far right, and in the appearing box, choose "Histogram".



In the next box, write or just highlight the input column and the bins that you just wrote. For output range, choose the place where you want the histogram to be, and check the box “Chart Output”. Also, check “Labels” to claim that we have column headings in the 1st row.

eg01-36titanic.xlsx [Compatibility Mode] - Excel

FILE HOME INSERT PAGE LAYOUT FORMULAS DATA REVIEW VIEW PAMFAX

From Access From Web From Text From Other Sources Existing Connections Refresh All Properties Edit Links Connections Sort Filter Clear Reapply Advanced Text to Columns Flash Fill Remove Duplicates Validation Data Consolidate What-If Analysis Relationships Group Ungroup Subtotal Outline Analysis

F1 : X ✓ fx Minimum age

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	pclass	survived	gender	age		Minimum age	0.167								
2	1	0	female	2		Maximum age	80								
3	1	0	male	30											
4	1	0	female	25											
5	1	0	male	39		Bins									
6	1	0	male	71		0									
7	1	0	male	47		10									
8	1	0	male			20									
9	1	0	male	24		30									
10	1	0	male	36		40									
11	1	0	male	25		50									
12	1	0	male	45		60									
13	1	0	male	42		70									
14	1	0	male	41		80									
15	1	0	male	48											
16	1	0	male												
17	1	0	male	45											
18	1	0	male												

Sheet1 Sheet2 Sheet3

READY Screenshots Chapter 1 eg01-36titanic.xls W 1 - Graphical displ... 11:27 PM 9/5/2015

The resulting histogram shows a unimodal and almost symmetric distribution of passenger ages on the Titanic.

eg01-36titanic.xlsx [Compatibility Mode] - Excel

FILE HOME INSERT PAGE LAYOUT FORMULAS DATA REVIEW VIEW PAMFAX CHART TOOLS DESIGN FORMAT

From Access From Web From Text From Other Sources Existing Connections Refresh All Properties Edit Links Connections Sort Filter Clear Reapply Advanced Text to Columns Flash Fill Remove Duplicates Validation Data Consolidate What-If Analysis Relationships Group Ungroup Subtotal Outline Analysis

Chart 1 : X ✓ fx

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	pclass	survived	gender	age		Minimum age	0.167		Bins	Frequency									
2	1	0	female	2		Maximum age	80		0	0									
3	1	0	male	30					10	86									
4	1	0	female	25					20	162									
5	1	0	male	39		Bins			30	361									
6	1	0	male	71		0			40	210									
7	1	0	male	47		10			50	132									
8	1	0	male			20			60	62									
9	1	0	male	24		30			70	27									
10	1	0	male	36		40			80	6									
11	1	0	male	25		50			More	0									
12	1	0	male	45		60													
13	1	0	male	42		70													
14	1	0	male	41		80													
15	1	0	male	48															
16	1	0	male																
17	1	0	male	45															
18	1	0	male																

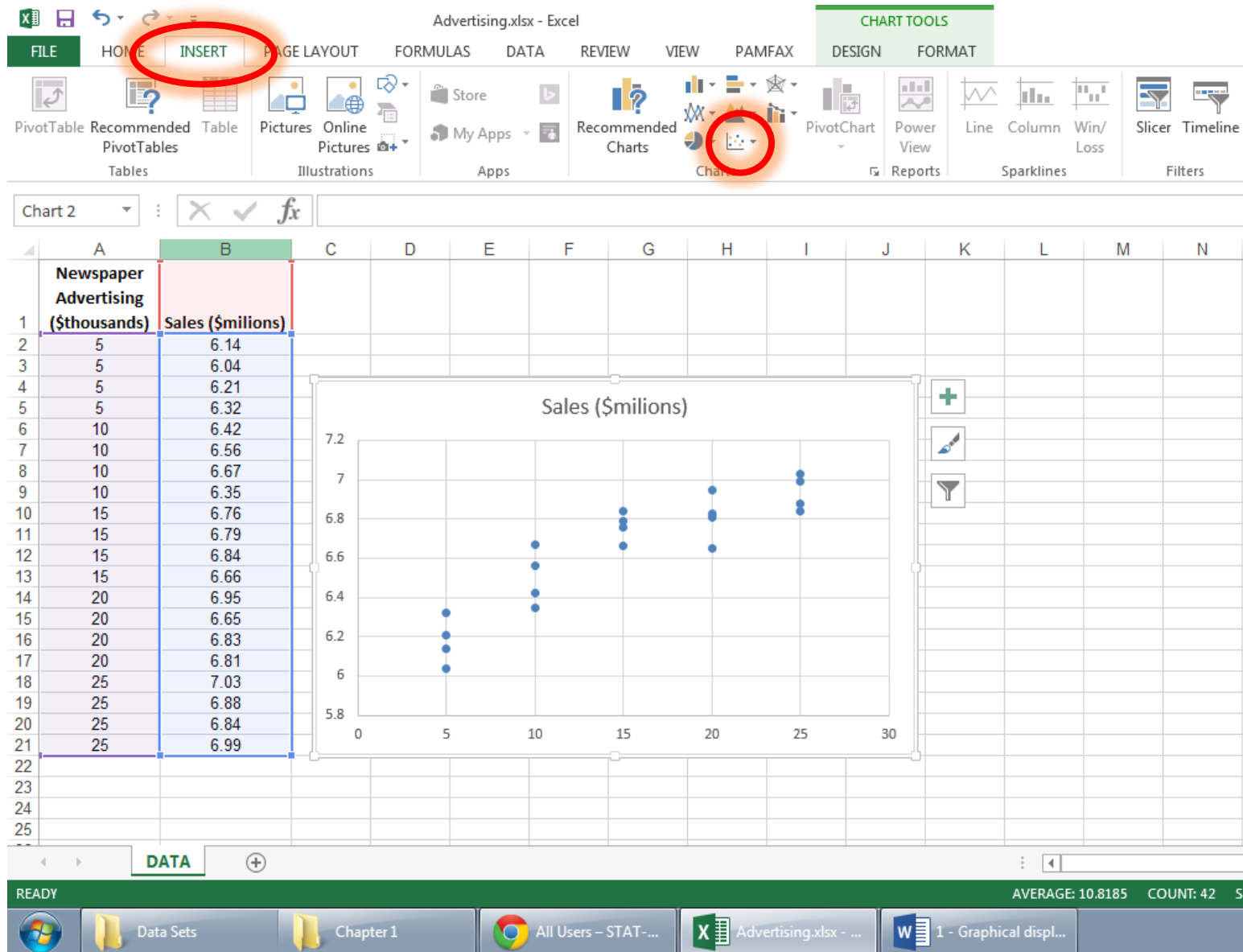
Sheet1 Sheet2 Sheet3

READY AVERAGE: 104.6 COUNT: 11 SUM: 1046 Screenshots Chapter 1 eg01-36titanic.xls W 1 - Graphical displ... 11:28 PM 9/5/2015

Scatterplots and Time Plots

Insert -> Charts -> Choose a scatterplot that you like.

The scatterplot below shows that on the average, the more a newspaper spends on advertising the larger sales it gets.



Time series plots are done similarly, with X-variable being time.